

Elektrodynamik Formelsamling

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1 Math

1.1 Vectors

$$a(\bar{A} + \bar{B}) = a\bar{A} + a\bar{B}$$

$$\bar{A} \cdot \bar{B} = |\bar{A}||\bar{B}|\cos(\theta) \quad (1.1)$$

$$\bar{A} \times \bar{B} = |\bar{A}||\bar{B}|\sin(\theta)\hat{n} \quad (1.4)$$

$$(\bar{A} \times \bar{B}) = -(\bar{B} \times \bar{A}) \quad (1.6)$$

$$\bar{A} \cdot (\bar{B} \times \bar{C}) = \bar{B} \cdot (\bar{C} \times \bar{A}) = \bar{C} \cdot (\bar{A} \times \bar{B}) \quad (1.15)$$

$$\bar{A} \times (\bar{B} \times \bar{C}) = \bar{B}(\bar{A} \cdot \bar{C}) - \bar{C}(\bar{A} \cdot \bar{B}) \quad (1.17)$$

$$(\bar{A} \times \bar{B}) \cdot (\bar{C} \times \bar{D}) = (\bar{A} \cdot \bar{C})(\bar{B} \cdot \bar{D}) - (\bar{A} \cdot \bar{D})(\bar{B} \cdot \bar{C})$$

$$\bar{A} \times [\bar{B} \times (\bar{C} \times \bar{D})] = \bar{B}[\bar{A} \cdot (\bar{C} \times \bar{D})] - (\bar{A} \cdot \bar{B})(\bar{C} \cdot \bar{D}) \quad (1.18)$$

1.2 Calculus

$$dT = \left(\frac{\partial T}{\partial x} \hat{x} + \frac{\partial T}{\partial y} \hat{y} + \frac{\partial T}{\partial z} \hat{z} \right) \cdot (dx\hat{x} + dy\hat{y} + dz\hat{z})$$

$$= (\nabla T) \cdot d\vec{l}, \quad (1.35)$$

where the gradient (∇) is an operator:

$$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}. \quad (1.39)$$

The gradient is used to calculate the divergence of a vector with:

$$\nabla \cdot \bar{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}, \quad (1.40)$$

and the curl with:

$$\nabla \times \bar{v} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix}$$

$$= \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{x} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{y} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{z}. \quad (1.41)$$

The product rules for the gradient are:

$$\nabla(fg) = f\nabla g + g\nabla f$$

$$\nabla(\bar{A} \cdot \bar{B}) = \bar{A} \times (\nabla \times \bar{B}) + \bar{B} \times (\nabla \times \bar{A}) + (\bar{A} \cdot \nabla)\bar{B} + (\bar{B} \cdot \nabla)\bar{A}$$

$$\nabla \cdot (f\bar{A}) = f(\nabla \cdot \bar{A}) + \bar{A} \cdot (\nabla f)$$

$$\nabla \cdot (\bar{A} \times \bar{B}) = \bar{B} \cdot (\nabla \times \bar{A}) - \bar{A} \cdot (\nabla \times \bar{B})$$

$$\nabla \times (f\bar{A}) = f(\nabla \times \bar{A}) + \nabla f \times \bar{A}$$

$$\nabla \times (\bar{A} \times \bar{B}) = \bar{A}(\nabla \cdot \bar{B}) - \bar{B}(\nabla \cdot \bar{A}) + (\bar{B} \cdot \nabla)\bar{A} - (\bar{A} \cdot \nabla)\bar{B}$$

When we use the gradient to make a second derivative we get the Laplacian:

$$\nabla(\nabla T) = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}, \quad (1.42)$$

and the laplacian of a vector:

$$\nabla^2 \vec{v} = (\nabla^2 v_x) \hat{x} + (\nabla^2 v_y) \hat{y} + (\nabla^2 v_z) \hat{z}. \quad (1.43)$$

We also have the following identities:

$$\nabla \times (\nabla T) = 0 \quad (1.45)$$

$$\nabla \cdot (\nabla \times \vec{v}) = 0 \quad (1.46)$$

$$\nabla \times (\nabla \times \vec{v}) = \nabla(\nabla \cdot \vec{v}) - \nabla^2 \vec{v}. \quad (1.47)$$

Dirac Delta Function

$$\begin{aligned} \delta(x-a) &= \begin{cases} 0, & \text{if } x \neq a \\ \infty, & \text{if } x = a \end{cases} \\ \int_{-\infty}^{\infty} \delta(x-a) dx &= 1 \\ \int_{-\infty}^{\infty} f(x) \delta(x-a) dx &= f(a) \\ \int_{-\infty}^{\infty} f(x) \delta(g(x)) dx &= \sum_{x_i} \frac{f(x_i)}{|g'(x_i)|} \\ \delta(x^2 - a^2) &= \frac{\delta(x-a) + \delta(x+a)}{2|a|} \end{aligned}$$

1.3 Fundamental theorem of calculus

Gradient:

$$\int_{\vec{a}}^{\vec{b}} (\nabla T) \cdot d\vec{l} = T(\vec{b}) - T(\vec{a}) \quad (1.55)$$

Divergence:

$$\int_V (\nabla \cdot \vec{v}) dV = \oint_S \vec{v} \cdot d\vec{a} \quad (1.56)$$

Curl:

$$\int_S (\nabla \times \vec{v}) \cdot d\vec{a} = \oint_{\mathcal{P}} \vec{v} \cdot d\vec{l} \quad (1.57)$$

2 Electrostatics

2.1 Electric field

The force on a test charge (Q) from a single point charge (q), a distance $\bar{r}$ away, is calculated by Coulomb's law:

$$\bar{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{\bar{r}^2} \hat{r}, \quad (2.1)$$

where $\epsilon_0 = 8.85 \cdot 10^{-12} \frac{\text{C}^2}{\text{N}\cdot\text{m}^2}$ is the permittivity of free space.

If we have several point charges, at different distances from Q , we have an electric field:

$$\bar{F} = Q\bar{E}, \quad (2.3)$$

$$\bar{E} = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i}{\bar{r}_i^2} \hat{r}_i. \quad (2.4)$$

If we have a continuous charge distribution, we can calculate the electric field by:

$$\bar{E} = \frac{1}{4\pi\epsilon_0} \int \frac{1}{\bar{r}^2} \hat{r} dq, \quad (2.5)$$

where dq is the charge in a small element, be it line charge ($dq = \lambda dl$), area charge ($dq = \sigma da$) or volume charge ($dq = \rho d\tau$).

Special cases:

Field of infinite length wire:

$$E = \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{z}$$

Field of infinite plane:

$$\bar{E} = \frac{\sigma}{2\epsilon_0} \hat{a}$$

Field outside a uniformly charged sphere with total charge q , $r > R$:

$$\bar{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

2.2 Divergence and Curl

Field lines are imaginary lines used to represent the direction and strength of a field, such as an electric field. They start on positive charges and end on negative charges. The density of these lines indicates the field's strength; closer lines mean a stronger field.

Flux refers to the quantity of a field passing through a given surface. For an electric field, it is the product of the electric field strength and the area

through which the field lines pass, considering the angle between the field lines and the surface.

$$\Phi = \int_S \vec{E} \cdot d\vec{a} \quad (2.11)$$

Gauss's law states that the total electric flux through a closed surface is proportional to the charge enclosed within that surface.

$$\oint \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enc}}}{\epsilon_0}, \quad (2.13)$$

where Q_{enc} is the charge enclosed by the surface. It can be rewritten with the divergence theorem, when $Q_{\text{enc}} = \int_V \rho d\tau$:

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}. \quad (2.14)$$

To calculate the curl of the electric field we see that:

$$\int_{\vec{a}}^{\vec{b}} \vec{E} \cdot d\vec{l} = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r_a} - \frac{q}{r_b} \right), \quad (2.18)$$

which on a closed interval gives us:

$$\oint \vec{E} \cdot d\vec{l} = 0, \quad (2.19)$$

$$\nabla \times \vec{E} = \vec{0}. \quad (2.20)$$

2.3 Electric potential

Since we have $\oint \vec{E} \cdot d\vec{l}$, it must be independent of the path taken, and therefore we can calculate the potential of a point as:

$$V(\vec{r}) = - \int_{\mathcal{O}}^{\vec{r}} \vec{E} \cdot d\vec{l}, \quad (2.21)$$

and the electric potential difference between two points as:

$$V(\vec{b}) - V(\vec{a}) = - \int_{\vec{a}}^{\vec{b}} \vec{E} \cdot d\vec{l}, \quad (2.22)$$

$$\Rightarrow \vec{E} = -\nabla V. \quad (2.23)$$

Poisson's equation:

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}, \quad (2.24)$$

which gives us a way to calculate the potential from the charge distribution:

For a collection of point charges:

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}, \quad (2.27)$$

For a continuous charge distribution:

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{z} dq, \quad (2.28)$$

For a volume charge distribution:

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{z} d\tau'. \quad (2.29)$$

Special cases:

Potential of infinite length wire:

$$V = \frac{1}{4\pi\epsilon_0} \lambda \ln \left(\frac{b}{a} \right)$$

Potential of infinite plane:

$$V = \frac{\sigma}{2\epsilon_0} z$$

Potential of spherical shell:

$$V = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{q}{r} & r > R \\ \frac{1}{4\pi\epsilon_0} \frac{q}{R} & r < R \end{cases}$$

Figure (2.35) shows the way to go between electric field, electric potential and volume charge:

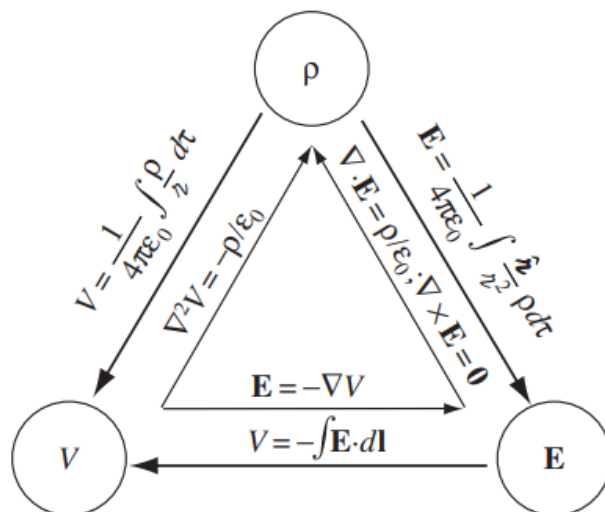


FIGURE 2.35

2.4 Work and Energy

If you want to move a test charge (Q) from point $\bar{a}$ to point $\bar{b}$, you have to do work against the electric field. The work done is the potential difference between the two points:

$$W = Q [V(\bar{b}) - V(\bar{a})]. \quad (2.38)$$

So if you want to move the point from infinity to a point $\bar{r}$, and you have the reference point in infinity:

$$W = QV(\bar{r}). \quad (2.39)$$

The work to move a collection of charges is:

$$W = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \sum_{j>i}^n \frac{q_i q_j}{r_{ij}}, \quad (2.40)$$

$$= \frac{1}{8\pi\epsilon_0} \sum_{i=1}^n \sum_{j \neq i}^n \frac{q_i q_j}{r_{ij}}, \quad (2.41)$$

$$= \frac{1}{2} \sum_{i=1}^n q_i V(\bar{r}_i). \quad (2.42)$$

For continuous charge distributions:

$$W = \frac{1}{2} \int V dq, \quad (2.43)$$

where again, dq is the charge in a small element, be it line charge ($dq = \lambda dl$), area charge ($dq = \sigma da$) or volume charge ($dq = \rho d\tau$).

If we take (2.43) for volume over all space, we get:

$$W = \frac{\epsilon_0}{2} \int E^2 d\tau. \quad (2.45)$$

Special cases

Energy of Q spread uniformly throughout sphere of radius R :

$$\begin{aligned} W_{tot} &= W_{out} + W_{in} \\ &= \frac{Q^2}{8\pi\epsilon_0 R} + \frac{3Q^2}{20\pi\epsilon_0 R} \end{aligned}$$

2.5 Conductors

A conductor is a material where the electrons are free to move around. The electric field inside a conductor is zero, and the charges are distributed on the surface, like metal. The electric field inside a conductor is zero, and when it

is placed in an electric field, the charges collect on the surface to create an electrical field inside the conductor, that cancels out the initial field.

$$\nabla \bar{E} = \frac{\rho}{\epsilon_0} = 0$$

Induced charges on a conductor are always on the surface, and the electric field is always perpendicular to the surface. For a conductor with a cavity, the electric field is zero inside the cavity, and the charges are on the surface of the cavity. Outside the conductor, the electric field is the same as if the conductor was not there.

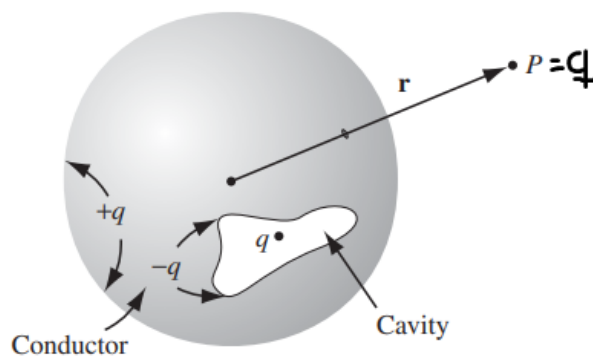


FIGURE 2.46

Figure 3.24 shows a conductor in an electric field, and how it would bend the field lines, so they are always perpendicular. so the electric field parallel to a conductor is 0.

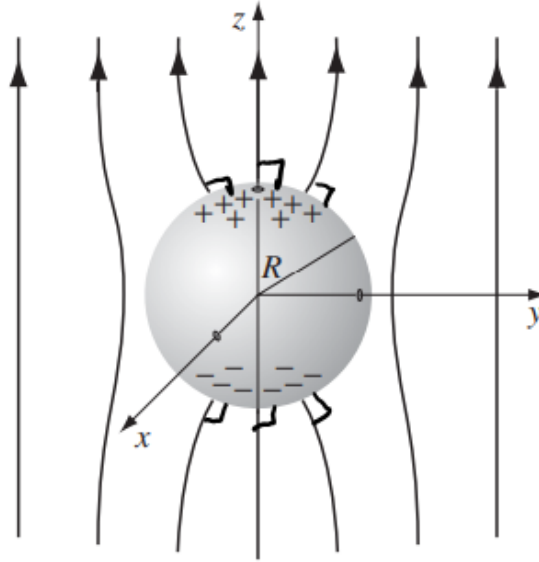


FIGURE 3.24

Because the field inside a conductor is zero, the field immediately outside is:

$$\vec{E} = \frac{\sigma}{\epsilon_0} \hat{n}, \quad (2.48)$$

The force per unit area and outward electrostatic pressure on the surface of a conductor is:

$$\vec{f} = \frac{1}{2\epsilon_0} \sigma^2 \hat{n}, \quad (2.51)$$

$$P = \frac{\epsilon_0}{2} E^2 = \frac{\sigma^2}{2\epsilon_0}. \quad (2.52)$$

Capacitors

If we have two conductors with charge Q and $-Q$, the electric potential between them is:

$$V = V_+ - V_- = - \int_{(-)}^{(+)} \vec{E} \cdot d\vec{l},$$

if the shape is complicated, $\vec{E}$ is hard to calculate, but we know it is proportional to Q , and therefore V is also proportional, which gives us the capacitance:

$$C = \frac{Q}{V}. \quad (2.53)$$

We can talk about the capacitance of a single capacitor, this refers to (2.53) where the reference point for V is infinity.

For a parallel-plate capacitor the capacitance is:

$$C = \frac{A\epsilon_0}{d}, \quad (2.54)$$

To charge a capacitor you have to move the charge from the positive plate to the negative plate. the energy required to do this is:

$$W = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C}. \quad (2.55)$$

3 Potential

3.1 Laplace's equation

To easily calculate the potential in a region, we can use Laplace's equation:

$$\nabla^2 V = 0. \quad (3.4)$$

In one dimension the solution is:

$$V(x) = ax + b, \quad (3.6)$$

but for 2 dimensions it is not easy to calculate exactly, but you can calculate $V(x, y)$ as the average of the potential around the point:

$$V(x, y) = \frac{1}{2\pi R} \oint_{circle} V dl,$$

which might make no sense, but it can be used to calculate the charge distribution, when boundary conditions are known.

In three dimensions the value of V at point $\bar{r}$ is the average value of V over a spherical surface of radius R centered at $\bar{r}$:

$$V(\bar{r}) = \frac{1}{4\pi R^2} \oint_{sphere} V da.$$

First uniqueness theorem: The solution to Laplace's equation in some volume $\mathcal{V}$ is uniquely determined if V is specified on the boundary surface $\mathcal{S}$. Meaning that it doesn't matter how you get your solution, if it satisfies Laplace's equation, and the boundary values are correct, then the answer is correct.

Second uniqueness theorem: In a volume $\mathcal{V}$ surrounded by conductors and containing a specified charge density ρ , the electric field is uniquely determined if the *total charge* on each conductor is given. (The region as a whole can be bounded by another conductor, or else unbounded.)

3.2 The Method of Images

The method of images can be used to solve advanced electrostatic problems, by replacing conductors with fictitious charges that mimic the same boundary conditions.

For example, we have a infinite grounded plate, with a point charge a distance d above it. The boundary potential is that $V = 0$ when $z = 0$ and that $V \rightarrow 0$ far from the charge (Fig 3.10). The same boundary conditions are satisfied if we have a point charge at $-d$ instead of the conductor (Fig 3.11). Therefore the potential for a pointcharge above an infinite grounded plate is:

$$V(x, y, z) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + (z - d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z + d)^2}} \right]. \quad (3.9)$$

From this we can calculate the induced surface charge on the conductor:

$$\begin{aligned}\sigma(x, y) &= -\epsilon_0 \frac{\partial V}{\partial z} \Big|_{z=0} \\ &= \frac{-qd}{2\pi(x^2 + y^2 + d^2)^{3/2}},\end{aligned}\tag{3.10}$$

and the total charge:

$$Q = \int \sigma da = -q.\tag{3.11}$$

We can also use the method of images to find the force on the point charge due to the induction:

$$\bar{F} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2d)^2} \hat{z},\tag{3.12}$$

but the work will not be the same for the plane, and the point charge, so if we want to calculate the energy to move the charge from infinity to d as:

$$W = \int_{\infty}^d \bar{F} \cdot d\bar{l} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d}.\tag{3.14}$$

You can not put the image charge in the region where you are calculating the potential, so in example 3.2, where we have a point charge a distance a away from a spherical grounded conductor, the image charge is inside the sphere.

This gives the potential:

$$V(\bar{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{\tau} + \frac{q'}{\tau'} \right),\tag{3.17}$$

where q' is the image charge, and τ and τ' is the distance from q and q' .

3.3 Seperation of variables

Chapter 3.3.1 shows and 3 exampls of how to calculate the potential between 2 parallel grounded conductors, connected by insulators. To calculate the potential between the plates we use separation of variables, so we asume the potential can be written as:

$$V(x, y) = X(x)Y(y)\tag{3.22}$$

$$\Rightarrow \frac{1}{X} \frac{\partial^2 X}{\partial x^2} + \frac{1}{Y} \frac{\partial^2 Y}{\partial y^2} = 0\tag{3.23}$$

which we can then solve from the boundary conditions of the problem, with the help of Fouriers trick.

In chapter 3.3.2 we find a genrerall soultion for the potential in sperical coordinates, with symmetry in φ :

$$V(r, \theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos(\theta)),\tag{3.65}$$

where A_l and B_l are constants determined by the boundary conditions, and $P_l(\cos(\theta))$ is the Legendre polynomials given by:

$$P_l(x) = \frac{1}{2^l l!} \left(\frac{d}{dx} \right)^l (x^2 - 1)^l. \quad (3.62)$$

$P_0(x) = 1$
$P_1(x) = x$
$P_2(x) = (3x^2 - 1)/2$
$P_3(x) = (5x^3 - 3x)/2$
$P_4(x) = (35x^4 - 30x^2 + 3)/8$
$P_5(x) = (63x^5 - 70x^3 + 15x)/8$

TABLE 3.1 Legendre Polynomials.

3.4 Multipole Expansion

A dipole is two opposite charges particles, collect two dipoles to make an quadropole, and so on.

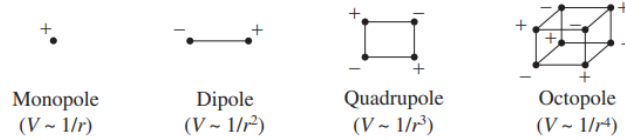


Fig. 3.27

To calculate the potential from a multipole we use:

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \int (r')^n P_n(\cos(\alpha)) \rho(\vec{r}') d\tau', \quad (3.95)$$

where P_l is the Legendre polynomial, ρ is the total charge of the distribution. $n = 0$ for monopole, 1 for dipole and so on.

For monopole and dipole we get:

$$V_{mon}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}, \quad (3.97)$$

$$V_{dip}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}, \quad (3.99)$$

where $\bar{p}$ is the dipole moment of the distribution:

$$\bar{p} = \int \bar{r}' \rho(\bar{r}') d\tau'. \quad (3.98)$$

The electric field of a dipole is:

$$\bar{E}_{dip}(r, \theta) = \frac{p}{4\pi\epsilon_0 r^3} (2 \cos(\theta) \hat{r} + \sin(\theta) \hat{\theta}). \quad (3.103)$$